

Ethan Yang

e99yang@uwaterloo.ca | 778-798-0213 | Richmond, BC | [LinkedIn](#) | [Portfolio](#) | [GitHub](#)

EDUCATION

University of Waterloo

Bachelor of Science, Honours Mathematical Physics

Waterloo, ON

Expected May 2030

PROJECTS

DeltaV - Climbing App

Jun 2026 – Present

UI/UX Design & Front-End Architecture

- Design and prototype mobile interface layouts using HTML, utilizing Claude Code and Google AI Studio to accelerate component generation before production deployment in SwiftUI.
- Structure the core application architecture and Supabase database backend, leveraging AI tools to automate data schemas and transition components toward a SwiftUI production roadmap.
- Collaborate with a partner to integrate real-time sensor data from an Arduino-based IMU component, translating raw climbing telemetry into a 3-D rendering

Personal Portfolio Website

Jun 2026 – Present

Web Development

- Architect a responsive personal portfolio using HTML, CSS, and TypeScript, utilizing advanced prompt engineering via Claude Code to direct UI/UX refinements and accelerate feature deployment by 80%

VEX Robotics PID & Odometry System – Team 604X

Sept 2023 – May 2024

Control Systems, Programming, and Troubleshooting

- Deployed a closed-loop C++ PID control and kinematic odometry system to track robot position in real time, improving autonomous path-tracking accuracy and minimizing steady-state error.

EXPERIENCE

Western Pacific Robotics Academy (WPRa)

Richmond, BC

Robotics Instructor

Sept 2023 – Present

- Administer workshop logistics and maintenance, maintain accurate inventory records, and provide front-line customer service to parents and students.
- Instruct students on core VEX robotics concepts, including CAD design, C++ programming, and control system troubleshooting.
- Facilitate the academy's growth as an instructor, achieving a 60%+ increase in active membership.

Team Captain & Mechanical Lead – VEX Teams 604X and 886W

Sept 2019 – May 2025

- Led the mechanical design, prototyping, and construction of VEX V5 robots, contributing to 40+ recognitions for excellence in robot design and competition performance.
- Helped scale WPRa into a top-ranked VEX Robotics academy in British Columbia, with more than 15 active teams.
- Captained teams to qualify for the VEX World Championship each season, including a 2020 Division finalist finish (top 0.4% of 20,000+ teams), while coordinating a multidisciplinary team under tight deadlines.

University of Waterloo Formula Electric (UWFE)

Waterloo, ON

Chassis Team Member

Sept 2025 – Present

- Design, prototype, and fabricate structural chassis components using CAD software, machining processes, and composite manufacturing techniques to ensure reliability and performance.
- Coordinate with leads and responsible engineers across multiple subsystems to implement performance-driven design improvements.

KEY SKILLS

Software: C++ (Basic), Python (Basic), Microsoft 365, Visual Studio Code, GitHub

Engineering Design: Fusion 360, Onshape, Composite Manufacturing, Machine Tools, 3D Printing

Web Development: HTML (Basic), CSS, TypeScript, Claude Code (AI-Assisted Dev), Prompt Engineering

Languages: English (Fluent), Mandarin (Conversational), Spanish (Elementary)